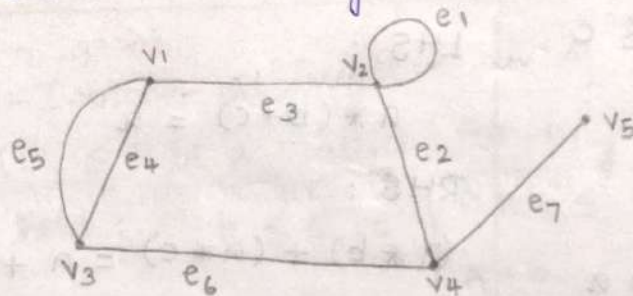


# MODULE - 6 FUNDAMENTALS OF GRAPHS

## Graph :

A graph  $G$  consists of a set of objects  $V = \{v_1, v_2, v_3, \dots\}$  called vertices (also called points / nodes) and other set  $E = \{e_1, e_2, e_3, \dots\}$  whose elements are called edges (also called lines / arcs).

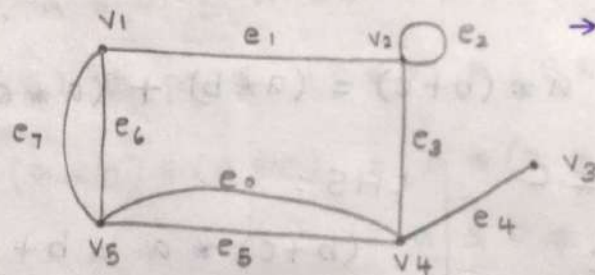
The set  $V(G)$  is called vertex set of  $G$  &  $E(G)$  is the edge set. It is usually denoted as  $G = (V, E)$ .



→ A graph with 5 vertices & 7 edges

→  $e_2, e_6$  &  $e_7$  are incident with vertex  $v_4$

→ A graph with  $p$ -vertices &  $q$ -edges is called a  $(p, q)$  graph. Eg:



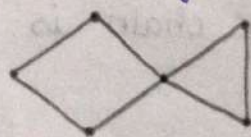
→ A graph with 5 vertices & 8 edges is called a  $(5, 8)$  graph.

→ The  $(1, 0)$  graph is trivial graph.

## Basic terminologies :

**Loop :** An edge of a graph that joins a node to itself is called loop / self loop. i.e, a loop is an edge  $(v_i, v_j)$  where  $v_i = v_j$ .

**Simple graph :** A graph which has neither loops nor multiple edges.

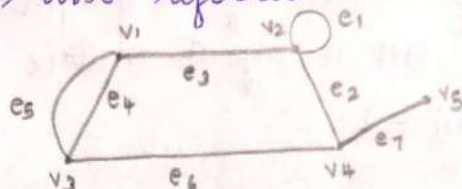


→ simple graph

**Degree of a vertex :** The no. of edges incident on a vertex  $v_i$  with self-loops counted twice is called the degree of vertex  $v_i$  and is denoted by  $\deg_G(v_i) / \deg v_i / d(v_i)$ .

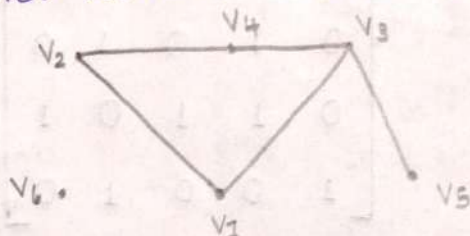


→ also referred to as its valency.



$$d(v_3) = d(v_4) = 3, d(v_5) = 1, d(v_2) = 4.$$

**Isolated vertex:** A vertex having no incident edge. In other words, vertex with zero degree.



Here,  $v_6$  is the isolated vertex.

**Pendent / end vertex:** A vertex of degree one. In above diagram,  $v_5$  - pendent vertex.

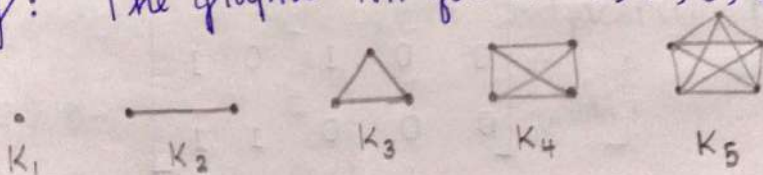
**Null graph:** A graph which contains only isolated node. It is denoted by  $N_n$  for  $n$  vertices.

→ denoted by  $K_n$

**Complete graph:** A simple graph with each vertex in  $G$  is connected with every other vertex.

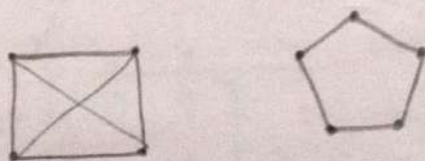
→  $K_n$  exactly has  $\frac{n(n-1)}{2}$  edges.

Eg: The graphs  $K_n$  for  $n = 1, 2, 3, 4, 5$



**Regular graph:** A graph in which all vertices are of equal degree. The complete graph  $K_n$  is a regular graph.

Eg:



**Note:**

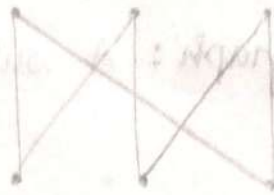
- The sum of degrees of the vertices in an undirected graph is even.
- The no. of vertices of odd degree in a graph is always even.

→ some more on last

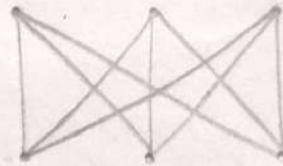
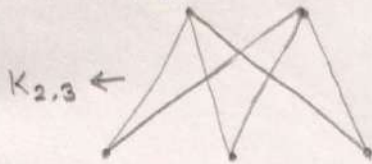


**Bipartite graph:** A graph whose vertex set can be partitioned into 2 subsets  $V_1$  &  $V_2$  such that each edge has one end in  $V_1$  & another end in  $V_2$ . Such partition  $(V_1, V_2)$  is called a bipartition of  $G$ .

Eg:

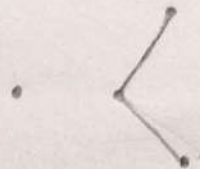
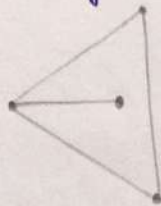


**Complete bipartite graph:** A simple bipartite graph with partition  $(V_1, V_2)$  in which every vertex of  $V_1$  is joined to every vertex of  $V_2$ . If  $|V_1| = m$ ,  $|V_2| = n$ , then it is denoted by  $K_{m,n}$ .



**Complement of  $G$ :** It is the simple graph with the same vertex set  $V$  in which 2 vertices are adj. if & only if they are not adj. in  $G$ . It is denoted by  $G_c$ .

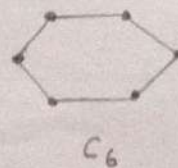
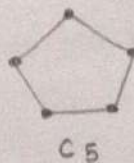
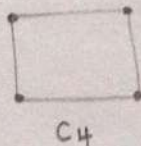
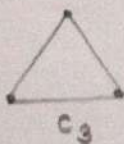
Eg:



**Degree sequence of  $G$ :** Let  $G$  be any graph with  $V(G) = V_1, V_2, \dots, V_V$ . Then the seq.  $d(V_1), d(V_2), \dots, d(V_V)$  is called degree sequence of  $G$ .

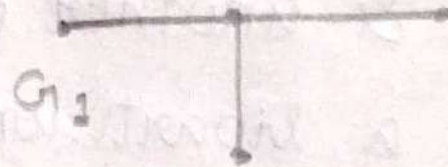
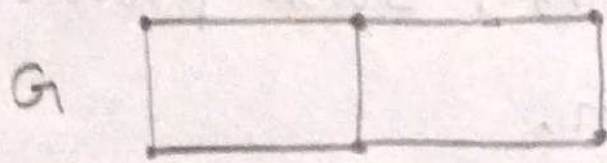
**Cycles:** The cycle  $C_n$ ,  $n \geq 3$

Eg:



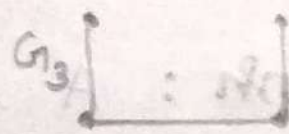
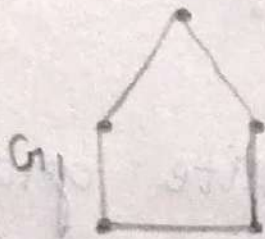
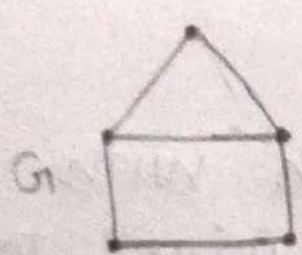


**Sub graph:** A graph having all of its / some / none of the edges & vertices of original graph. It is a subset of another graph.



→  $G_1$  is a sub-graph of  $G$ .

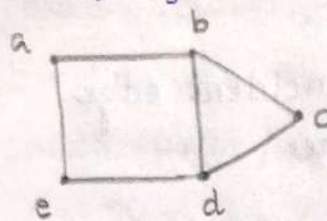
**Spanning sub graph:** A subgraph containing all the vertices of  $G$ .



$G_1$  &  $G_2$  - spanning subgraph of  $G$ .

$G_3$  - not a spanning subgraph of  $G$ .

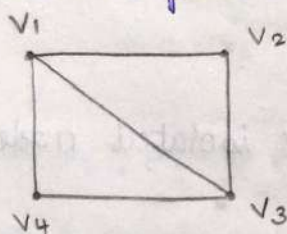
**Adjacency matrix :** The adjacency matrix of a graph  $G$  with  $n$  vertices & no  $11^2$  edges is an  $n$  by  $n$  matrix  $A = \{a_{ij}\}$  whose elts. are gn. by  $a_{ij} = 1$ .



$$A(a) = \begin{matrix} & \begin{matrix} a & b & c & d & e \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

**Problems :**

1. Use adjacency matrix to represent the graph.



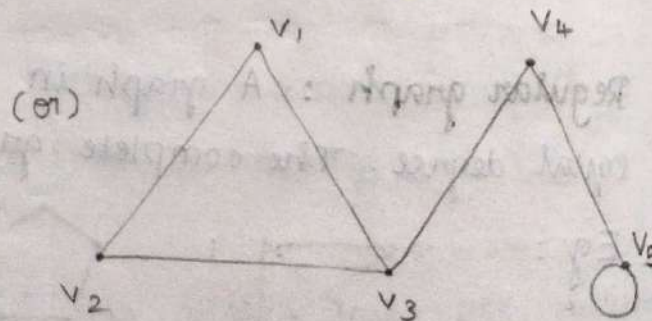
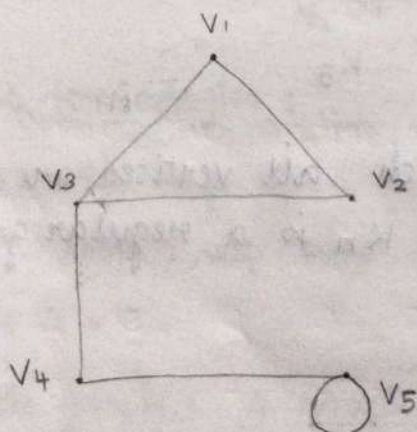
$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

2. Draw the undirected graph represented by adjacency matrix  $A$  given by

i)

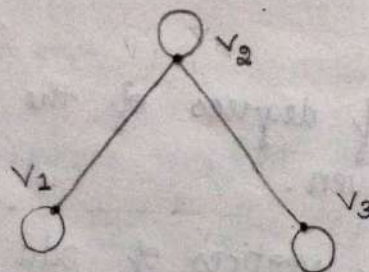
$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

Soln :



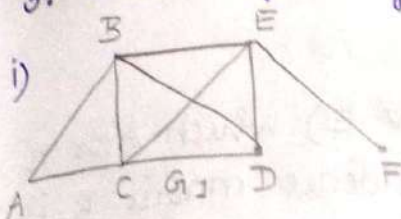
ii)

$$\begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

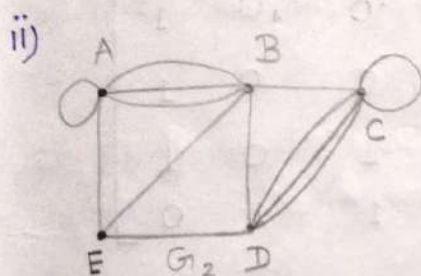




3. Use adjacency matrix to represent the matrix:



$$A = \begin{matrix} & \begin{matrix} A & B & C & D & E & F \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \\ F \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$



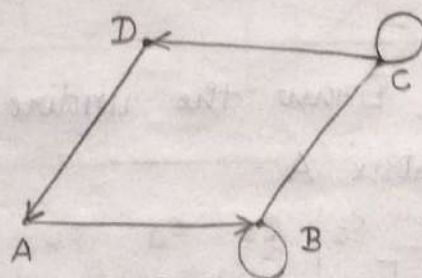
$$A = \begin{matrix} & \begin{matrix} A & B & C & D & E \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix} & \begin{bmatrix} 1 & 3 & 0 & 0 & 1 \\ 3 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 1 & 3 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

4. Construct graph from adjacency matrix:

i)

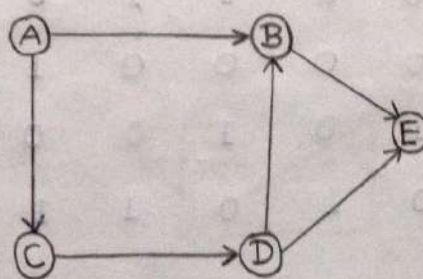
$$A = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Soln :



ii)

$$A = \begin{matrix} & \begin{matrix} A & B & C & D & E \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

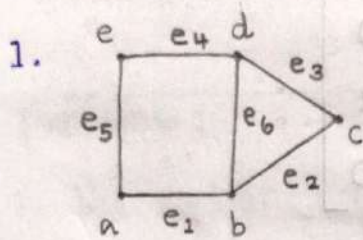


## Incidence matrix :

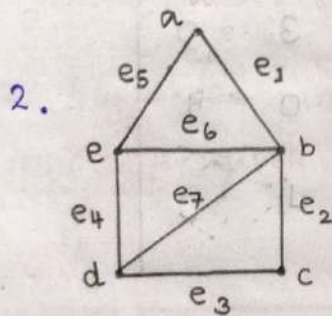
Consider a undirected graph  $G = (V, E)$  which has  $n$  vertices &  $m$  edges all labelled. The incidence matrix  $B = \{b_{ij}\}$  is then  $n \times m$  matrix,

where  $b_{ij} = 1$  when edge  $e_j$  is incident with  $v_i$ ,  
 $= 0$  otherwise

[Each column will have only 2 ones]  
 A 1 edge will have only 2 vertices



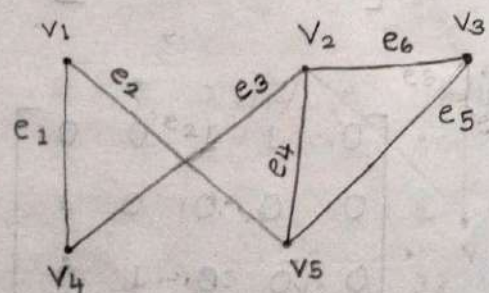
|   | $e_1$ | $e_2$ | $e_3$ | $e_4$ | $e_5$ | $e_6$ |
|---|-------|-------|-------|-------|-------|-------|
| a | 1     | 0     | 0     | 0     | 1     | 0     |
| b | 1     | 1     | 0     | 0     | 0     | 1     |
| c | 0     | 1     | 1     | 0     | 0     | 1     |
| d | 0     | 0     | 1     | 1     | 0     | 1     |
| e | 0     | 0     | 0     | 1     | 1     | 0     |



|   | $e_1$ | $e_2$ | $e_3$ | $e_4$ | $e_5$ | $e_6$ | $e_7$ |
|---|-------|-------|-------|-------|-------|-------|-------|
| a | 1     | 0     | 0     | 0     | 1     | 0     | 0     |
| b | 1     | 1     | 0     | 0     | 0     | 1     | 1     |
| c | 0     | 1     | 1     | 0     | 0     | 0     | 0     |
| d | 0     | 0     | 1     | 1     | 0     | 0     | 1     |
| e | 0     | 0     | 0     | 1     | 1     | 1     | 0     |

3. Draw the undirected graph by the gn. incidence matrix A

|       | $e_1$ | $e_2$ | $e_3$ | $e_4$ | $e_5$ | $e_6$ |
|-------|-------|-------|-------|-------|-------|-------|
| $v_1$ | 1     | 1     | 0     | 0     | 0     | 0     |
| $v_2$ | 0     | 0     | 1     | 1     | 0     | 1     |
| $v_3$ | 0     | 0     | 0     | 0     | 1     | 1     |
| $v_4$ | 1     | 0     | 1     | 0     | 0     | 0     |
| $v_5$ | 0     | 1     | 0     | 1     | 1     | 0     |





## Graphs isomorphism :

Let  $G_1 = (V_1, E_1)$  &  $G_2 = (V_2, E_2)$  be two graphs.

A function  $f: V_1 \rightarrow V_2$  is called a graphs isomorphism

if i)  $f$  is one-to-one & onto

ii) for all  $a, b \in V_1$ ,  $\{a, b\} \in E_1$  if & only if  $\{f(a), f(b)\} \in E_2$  when such function exists,  $G_1$  &  $G_2$  are called isomorphic graphs & is written as  $G_1 \cong G_2$ .

Necessary conditions for 2 graphs to be isomorphic are

Both must have

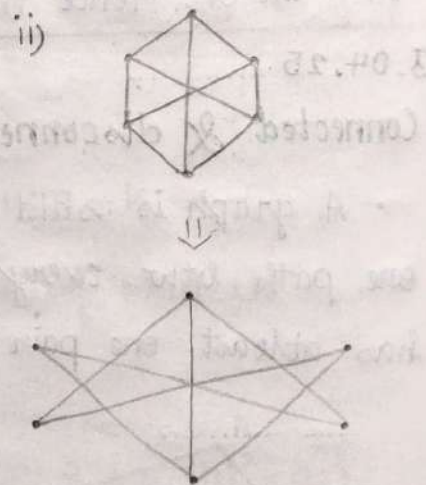
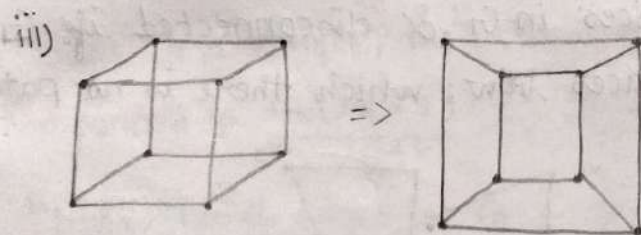
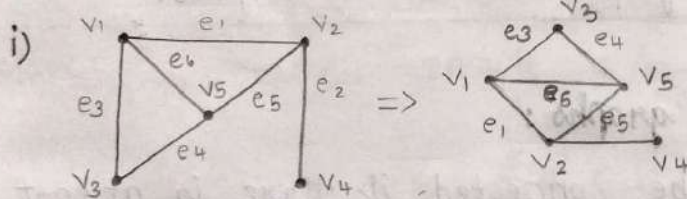
i) Same no. of vertices.

ii) Same no. of edges.

iii) Equal no. of vertices with same degree.

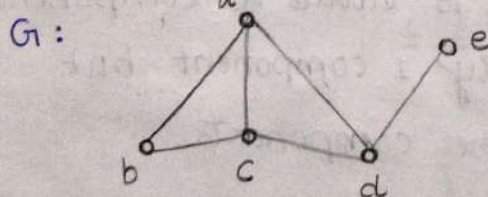
iv) Same degree sequence & same cycle vector  $(c_1, \dots, c_n)$ , where  $c_i$  is the no. of cycles of length  $i$ .

Examples :



Problems :

1. Show that the following graphs are isomorphic.

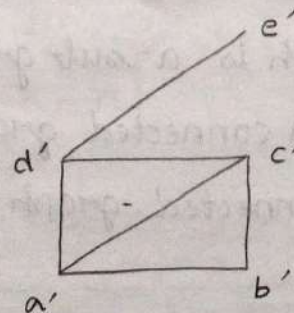


$\deg(G)$

$\deg(a) = 3$

$\deg(b) = 2$

$G'$ :



$\deg(G')$

$\deg(a') = 3$

$\deg(b') = 2$



$$\deg(c) = 3$$

$$\deg(c') = 3$$

$$\deg(d) = 3$$

$$\deg(d') = 3$$

$$\deg(e) = 1$$

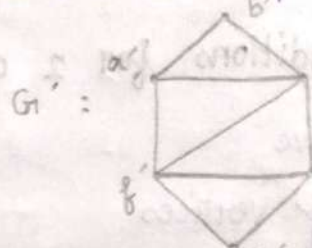
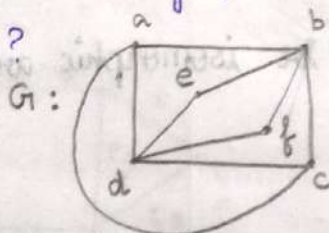
$$\deg(e') = 1$$

Both  $G$  &  $G'$  has 5-vertices & 6-edges

Hence the correspondance is  $a-a'$ ,  $b-b'$ , ...,  $e-e'$ .

$\therefore$  the gn. 2 graphs are isomorphic.

2. Are the 2-graphs, gn. below, is isomorphic? Give a reason?



Vertices of degree 4 :  $b-f'$   
 $d-c'$

Vertices of degree 3 :  $a-a'$   
 $c-d'$

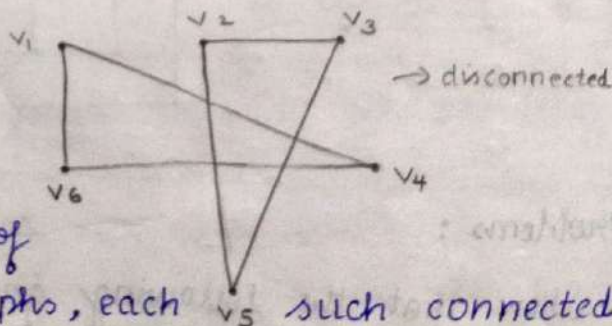
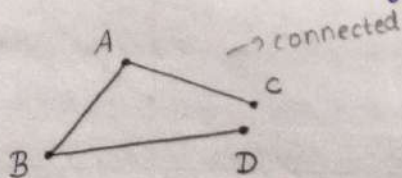
Vertices of degree 2 :  $e-b'$   
 $f-e'$

Now the vertices of deg. 3, in  $G$  are  $a$  &  $c$  & they are adjacent in  $G'$ , while these are  $a'$  &  $d'$  which are not adj in  $G$ . Hence the 2 graphs are not isomorphic.

3.04.25

Connected & disconnected graphs :

• A graph is said to be connected if there is atleast one path btw. every 2 vertices in  $G$  & disconnected if  $G$  has atleast one pair of vertices btw. which there is no path.



• Every graph  $G$  consists of one or more connected graphs, each such connected graph is a sub graph of  $G$  & is called a component of  $G$ . A connected graph has only 1 component but disconnected graph has 2/more components.



Theorem :

A simple graph with  $n$  vertices and  $k$  components can have at most  $\frac{(n-k)(n-k+1)}{2}$  edges.

Proof :

Let  $n_1, n_2, n_3, \dots, n_k$  be the number of vertices in each of  $k$  components of the graph  $G$ .

Then  $n_1 + n_2 + n_3 + \dots + n_k = n = V(G)$  i.e.,  $\sum_{i=1}^k n_i = n$

$$\sum_{i=1}^k (n_i - 1) = (n_1 - 1) + (n_2 - 1) + \dots + (n_k - 1) = \sum_{i=1}^k n_i - k = n - k$$

Squaring on b.s.,

$$(n_1 - 1)^2 + (n_2 - 1)^2 + \dots + (n_k - 1)^2 \leq (n - k)^2$$

$$[n_1^2 - 2n_1 + 1 + n_2^2 - 2n_2 + 1 + \dots + n_k^2 - 2n_k + 1] \leq (n - k)^2$$

$$\sum_{i=1}^k n_i^2 - 2 \sum_{i=1}^k n_i + k \leq (n - k)^2$$

$$\sum_{i=1}^k n_i^2 - 2 \sum_{i=1}^k n_i + k \leq (n - k)^2$$

$$\sum_{i=1}^k n_i^2 - 2n + k \leq (n - k)^2$$

$$\sum_{i=1}^k n_i^2 \leq (n - k)^2 + 2n - k \quad \text{--- (1)}$$

Since  $G$  is simple, the max. no. of edges of  $G$  in its components is  $\frac{n_i(n_i - 1)}{2}$ .

$$\text{Max. no. of edges of } G = \sum_{i=1}^k \frac{n_i(n_i - 1)}{2} = \sum_{i=1}^k \frac{(n_i^2 - n_i)}{2}$$

$$\sum n_i^2 \leq n^2 + k^2 - 2nk + 2nk$$

$$= \frac{1}{2} \left[ \sum_{i=1}^k n_i^2 - \sum_{i=1}^k n_i \right]$$

$$\leq \frac{1}{2} \left[ (n - k)^2 + 2n - k - n \right]$$

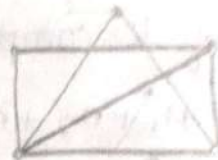
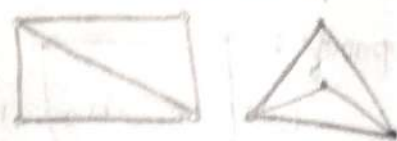
$$\leq \frac{1}{2} \left[ (n - k)^2 + (n - k) \right] \leq \frac{(n - k)(n - k + 1)}{2}$$

$$\text{Max. no. of edges of } G \leq \frac{(n - k)(n - k + 1)}{2}$$



**Planar graphs** : A graph is said to be a planar if there exists some geometric representation of  $G$  which can be drawn on a plane such that no 2 of its edges intersect. The point of intersection are called crossovers.

- A graph that cannot be drawn on a plane without a crossover b/w. its edges is called non-planar.

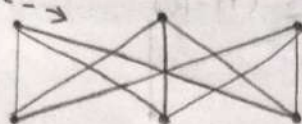


**Note :**

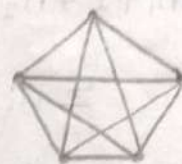
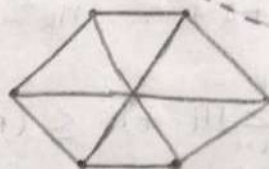
- The complete graph of 5 vertices is non-planar.



- $K_{3,3}$  &  $K_5$  are non-planar.



$\Rightarrow$



**Walk** : A finite alternative sequence of vertices / edges.

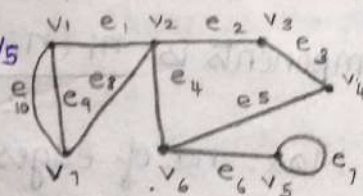
i) each edge in the seq. is incident on the vertices preceding & following it in the seq.

ii) no edge appears more than once in the seq.

• Also referred to as an edge train / a chain.

$\rightarrow v_2 e_4 v_6 e_5 v_4 e_3 v_3$  &  $v_7 e_8 v_2 e_4 v_6 e_6 v_5 e_7 v_5$

are walks



**Walk**

**Closed walk**

**Open walk**

$\rightarrow$  Begins & ends at same vertex

$\rightarrow$  A walk that is not closed / begins & ends at different vertex.

Eg:

$v_1 e_9 v_7 e_8 v_2 e_1 v_1$

Eg:

$v_5 e_7 v_5 e_6 v_6 e_5 v_4$

**Path** : An open walk in which no vertex appears more than once. (or simple / elementary path)

$v_6 e_5 v_4 e_3 v_3 e_2 v_2 \rightarrow$  path

$v_5 e_7 v_5 e_6 v_6 \rightarrow$  open walk but not path



**Circuit** : A closed walk in which no vertex appears more than once. (except initial & final vertex)

Eg:  $V_1 e_1 V_2 e_8 V_7 e_9 V_1$  &  $V_2 e_4 V_6 e_5 V_4 e_3 e_2 V_2 \rightarrow$  circuits

**Length** : No. of edges in a walk.

Open walk :  $V_6 e_6 V_5 e_7 V_5 = 2$

Closed walk :  $V_1 e_9 V_7 e_8 V_2 e_1 V_1 = 3$

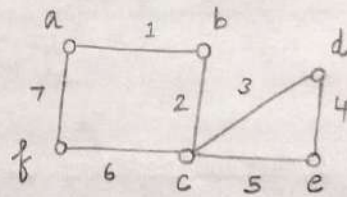
length

**Note :**

1. Edges only once - walk, path, circuits.
2. Vertex more than once in walk but not in path.
3. A path is an open walk, but an open walk need not to be a path.
4. A circuit is a closed walk, but a closed walk need not to be a circuit.

**Euler Graphs** : If a closed walk in a graph contains all the edges of the graph, then the walk is called an Euler line & graph is Euler graph.

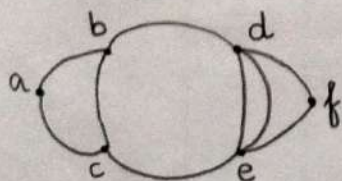
- The gn. graph is an Euler graph as  $a \rightarrow 1 \rightarrow b \rightarrow 2 \rightarrow c \rightarrow 3 \rightarrow d \rightarrow 4 \rightarrow e \rightarrow 5 \rightarrow c \rightarrow 6 \rightarrow f \rightarrow 7 \rightarrow a$  covers all the edges of the graph.



- A connected graph  $G$  is an Euler graph if & only if all the vertices of  $G$  are of even degree.
- A connected graph  $G$  is Eulerian if & only if its edge set can be decomposed into cycles.

**Euler path / Euler line :**

It includes every edges only once. Also called Euler trail.



$\rightarrow$  No Euler path & Euler circuit possible

**Euler circuit :**

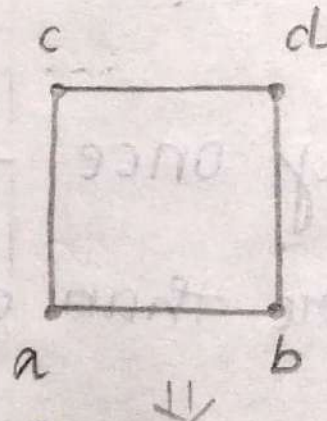
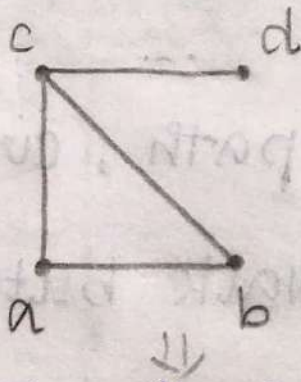
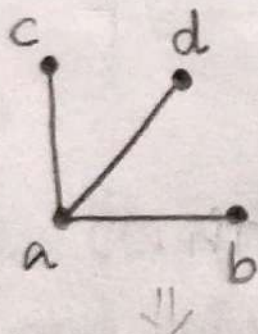
An Euler path that is circuit is called Euler circuit. A graph which has a Eulerian circuit is called an Eulerian graph.



**Hamiltonian circuit:** A closed walk that traverses every vertex of  $G$  exactly once, except of course the starting vertex at which the walk also terminates.

- A Hamiltonian circuit includes every vertex of  $G$ .
- It consists of  $n$  edges for  $n$  vertices.

**Hamiltonian path:** It is a simple path that contains all vertices of  $G$  where end pts. may be distinct.



X Hamiltonian path

✓ Hamiltonian path

X Hamiltonian cycle

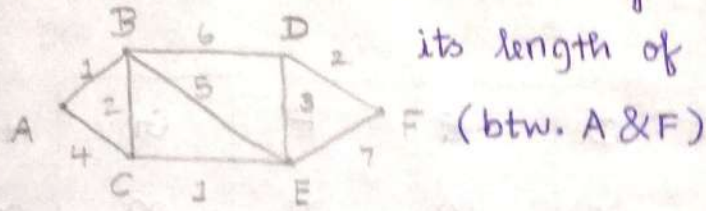
X Hamiltonian cycle

✓ Hamiltonian cycle



## Problems:

1. Use Dijkstra's algorithm to find the shortest path btw the indicated vertices in the weighted graph and also find its length of the shortest path.



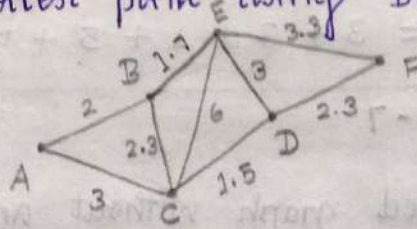
Soln.:

|   | A  | B                                                | C                                                | D                                                | E                                                | F                                                |
|---|----|--------------------------------------------------|--------------------------------------------------|--------------------------------------------------|--------------------------------------------------|--------------------------------------------------|
| A | 0A | 1A                                               | 4A                                               | $\infty$                                         | $\infty$                                         | $\infty$                                         |
| B | -  | <span style="border: 1px solid black;">1A</span> | 3B                                               | 7B                                               | 6B                                               | $\infty$                                         |
| C | -  | -                                                | <span style="border: 1px solid black;">3B</span> | 7B                                               | 4C                                               | $\infty$                                         |
| D | -  | -                                                | -                                                | <span style="border: 1px solid black;">7B</span> | <span style="border: 1px solid black;">4C</span> | 11E                                              |
| E | -  | -                                                | -                                                | -                                                | -                                                | <span style="border: 1px solid black;">9D</span> |
| F | -  | -                                                | -                                                | -                                                | -                                                | <span style="border: 1px solid black;">9D</span> |

Shortest path = A → B → D → F

Shortest length = 1 + 6 + 2 = 9

2. Find the shortest path using Dijkstra's algorithm btw. A & F.



Soln.:

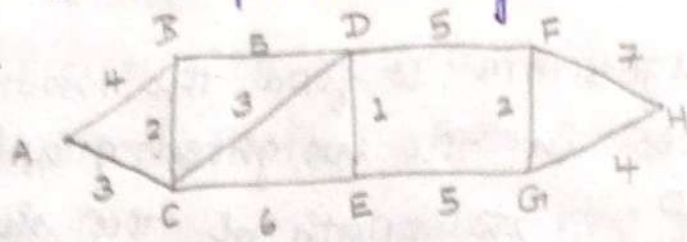
|   | A  | B                                                | C                                                | D                                                  | E                                                  | F                                                  |
|---|----|--------------------------------------------------|--------------------------------------------------|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------|
| A | 0A | 2A                                               | 3A                                               | $\infty$                                           | $\infty$                                           | $\infty$                                           |
| B | -  | <span style="border: 1px solid black;">2A</span> | 3A                                               | $\infty$                                           | 3.7B                                               | $\infty$                                           |
| C | -  | -                                                | <span style="border: 1px solid black;">3A</span> | 4.5C                                               | 3.7B                                               | $\infty$                                           |
| D | -  | -                                                | -                                                | <span style="border: 1px solid black;">4.5C</span> | <span style="border: 1px solid black;">3.7B</span> | 7E                                                 |
| E | -  | -                                                | -                                                | -                                                  | -                                                  | <span style="border: 1px solid black;">6.8D</span> |
| F | -  | -                                                | -                                                | -                                                  | -                                                  | <span style="border: 1px solid black;">6.8D</span> |

Length of shortest path = 6.8

Shortest path = A → C → D → F



3. Find the shortest path using Dijkstra algorithm bet. A & H.



|   | A                                                | B                                                | C                                                | D                                                | E                                                | F                                                 | G                                                 | H                                                 |
|---|--------------------------------------------------|--------------------------------------------------|--------------------------------------------------|--------------------------------------------------|--------------------------------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| A | <span style="border: 1px solid black;">0A</span> | 4A                                               | 3A                                               | $\infty$                                         | $\infty$                                         | $\infty$                                          | $\infty$                                          | $\infty$                                          |
| C | -                                                | 4A                                               | <span style="border: 1px solid black;">3A</span> | 6C                                               | 9C                                               | $\infty$                                          | $\infty$                                          | $\infty$                                          |
| B | -                                                | <span style="border: 1px solid black;">4A</span> | -                                                | 6C                                               | 9C                                               | $\infty$                                          | $\infty$                                          | $\infty$                                          |
| D | -                                                | -                                                | -                                                | <span style="border: 1px solid black;">6C</span> | 7D                                               | 11D                                               | $\infty$                                          | $\infty$                                          |
| E | -                                                | -                                                | -                                                | -                                                | <span style="border: 1px solid black;">7D</span> | 11D                                               | 12E                                               | $\infty$                                          |
| F | -                                                | -                                                | -                                                | -                                                | -                                                | <span style="border: 1px solid black;">11D</span> | 12E                                               | 18E                                               |
| G | -                                                | -                                                | -                                                | -                                                | -                                                | -                                                 | <span style="border: 1px solid black;">12E</span> | 16G                                               |
| H | -                                                | -                                                | -                                                | -                                                | -                                                | -                                                 | -                                                 | <span style="border: 1px solid black;">16G</span> |

Shortest path :  $A \rightarrow C \rightarrow D \rightarrow E \rightarrow G \rightarrow H$

Shortest distance =  $3 + 3 + 1 + 5 + 4 = 16$